Global Facilities - Design & Construction Standard - Mechanical - Process Cooling Water System

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1.0 Purpose

1.1 This document serves as a System Standard document that provides specifications that shall be followed during the Design, Engineering and Construction of new Process Cooling Water Systems as part of new Greenfield FAB Construction projects.

1.2 The document is intended to supplement the actual Construction Drawings and Construction Specifications for Construction projects.

2.0 Scope

2.1 The document covers the new Basebuild System installations only and does NOT cover tool installation.

2.2 Site(s) impacted

2.2.1 This document applies to all Micron Facilities employees, contractors, and vendors at all Micron sites for new Greenfield FAB Construction projects.

3.0 Definitions and Acronyms

- **CHW -** Chilled Water
- **FAT –** Factory Acceptance Test
- **FMCS/SCADA -** Facility Monitoring and Control System / Supervisory Control and Data Acquisition
- **FCT** – Facilities Central Team. This is used interchangeably with Global Facilities Engineering Team and associated to O&M BKMs
- **GFTT** – Global Facilities Technology Team. This is used interchangeably with Global Facilities Engineering Team and associated to design standards
- **GFC** – Global Facilities & Construction Team. This is used interchangeably with Global Facilities (GFac), Facilities Central Group (FCG), World Wide Operations (WWOps), World Wide Facilities (WW_FACILITIES), Corp Facilities
- **Global Facilities Central Team** – This is used interchangeably with Innovation & Integration Team
- **Global Facilities Construction Team** – This is used interchangeably with Global Construction Team
- **Global Facilities Operations Support Team** – This is used interchangeably with Global Facilities Operations Team
- **Global Facilities Planning Team** – This is used interchangeably with Global Facilities Line Planning Team
- **ICW -** Industrial City Water
- **IST –** Integrated System Test
- **ITP –** Inspection and Test Plan
- **LOTO/COHE -** Lockout Tagout / Control of Hazardous Energy
- **N+1+F -** System sizing concept for equipment quantity consisting of N duty units, +1 redundant standby unit, +F future unit(s)
- **OAT –** Operational Acceptance Test
- **PCW –** Process Cooling Water
- **PLC -** Programmable Logic Controller
- **POC/LPOC -** Point of Connection / Lateral Point of Connection
- **SAT –** Site Acceptance Test
- **TUM -** Tool Utility Matrix
- **UPS -** Uninterruptible Power Supply
- **UPW/DI/RO –** Ultrapure Water / De-Ionized Water / Reverse Osmosis Water
- **VSD/VFD -** Variable Speed Drive / Variable Frequency Drive

4.0 Roles and Responsibilities

Roles	Responsibilities
GFTT Lead (Discipline Engineer)	• Develop and maintain System Standard document • Ensure Best Known Methods (BKM), lessons learned, and emerging technologies are documented in System Standard • Perform periodic document review and update System Standard
FCT Lead (Discipline Engineer)	• Review and provide technical feedback on System Standard • Support the Site teams by providing knowledge and technical expertise
Site Project Construction Teams	• Review and provide technical feedback on System Standard • Incorporate System Standard into Project Engineering, Design, and Construction • Document project deviations, exceptions, and exclusions from System Standard • Share new lessons learned to GFTT Lead, to determine if it is Best Known Method (BKM) that can be captured in future revision of System Standard
Site Facilities Teams	• Use System Standard as technical reference

5.0 Overview of Systems

5.1 The Process Cooling Water System is a recirculating water system used to provide cooling water to manufacturing tools and equipment, including FAB Support vacuum pumps, standalone chillers, and other miscellaneous heat exchangers. Heat from the manufacturing tools and equipment is captured via the Process Cooling Water system, and then the heat is rejected to the central Chilled Water system.

5.2 The main Process Cooling Water equipment shall be as follows:
- 5.2.1 Distribution Pumps
- 5.2.2 Heat Exchangers
- 5.2.3 Filters

5.3 The auxiliary Process Cooling Water equipment and components shall be as follows:
- 5.3.1 Air Separator(s)
- 5.3.2 Expansion Tank(s)
- 5.3.3 Water Quality and Chemical Dosing Skid

6.0 Performance Spec Requirements:

6.1 The Process Cooling Water System shall be designed to achieve the operational performance specs as outlined in **Table 1**:

Table 1: Operational Performance Specs for PCW

Parameter	Specification Requirement	Remarks
Supply Temperature	15.56 ± 1.0 °C (60.0 ± 1.8 °F)	Measured at Main Supply Header After Heat Exchangers
Supply Pressure	0.60 ± 0.03 MPa (87 ± 4.3 psig)	Measured at Supply Lateral End of Line (at Subfab Level)
Conductivity	< 200 µS/cm (> 0.02 MΩ-cm)	Measured at sidestream return loop for Water Quality Sampling and Chemical Dosing

6.2 The Process Cooling Water System shall be designed to achieve the water quality specs as outlined in **Table 2**:

Table 2: Water Quality Specs for PCW

Parameter	Unit	Specification Requirement	Remarks
pH	-	7.0 – 9.0	
Conductivity	μS/cm	< 100 (> 0.01 MΩ-cm)	Measured at sidestream return loop for Water Quality Sampling and Chemical Dosing
M-alkalinity	mg $CaCO_3$ /l	<20	
Calcium hardness	mg $CaCO_3$ /l	<4	
Chloride Ion	mg/l as Cl^-	<100	
Iron	mg/l as Fe	<2	
Copper	mg/l as Cu	<0.3	
Turbidity	NTU	<5	
Corrosion Coupon, Cu	mdd	<2	
Corrosion Coupon, Al	mdd	<1	
Standard Plate Count bacteria	cfu/ml	<100	

7.0 Equipment Design and Sizing Requirements:

7.1 **System Type** – PCW system shall be "Closed" type system consisting of pressurized return to the main PCW system equipment.

7.2 **Redundancy and Future** – PCW system main equipment shall be designed for N+1+2F; for pumps, heat exchangers, and filters. All main system equipment shall have N+1 redundancy (with one redundant unit having capability as online standby), and a provision to install two additional future units (2F) for future capacity expansion.

7.3 **System Diversity and Sizing Factor** – PCW systems shall be sized so that the N capacity meets 115-120% of the PCW demand at full build (after initial construction of the system), based on the manufacturing tool layout and TUM. Therefore, the PCW system sizing for N capacity shall have 15-20% spare future capacity over the PCW demand at full build (after initial construction of the system).

7.4 To determine the PCW demand for N at full build, a diversity factor of 100% of the Peak demand from the TUM. In addition, the PCW system shall be sized to include the capacity of two additional future units (2F). See **Table 3**.

Table 3: PCW System Sizing

- PCW System Demand at Full Build,
  D = TUM Peak Demand x 100% Diversity
- PCW System N capacity,
  N = 115-120% x D
- PCW System Sizing = N + 2F

7.5 **Equipment Quantity** – PCW System main equipment shall be selected for a minimum of 3 units (2+1) or more, with each unit to be equal in capacity sizing; for Pumps, Heat Exchangers, and Filters.

7.6 **Equipment Arrangement** – All Main PCW equipment shall be arranged in parallel together and connected via a common piping header for each equipment set; for Pumps, Heat Exchangers, and Filters. This allows any combination of equipment to be in operation concurrently.

7.7 **Equipment Layout** – Equipment shall be suitably located to comply with all equipment maintenance clearances and access requirements. Adequate space and a clear routing path shall be reserved for future move-in of new equipment and to allow move-in and move-out of large components during maintenance and repair (such as motors, filters etc.).

7.8 Equipment Selection Criteria - See Table 4.

Table 4: Equipment Selection Criteria

Equipment	Type and Description
Distribution Pumps	• Pumps shall be Centrifugal, End-Suction Type or Split-Case Type. • Pump sizing and selection shall be as follows: ○ Pumps shall be selected with the performance indicated at the design operating point (flow and pressure, at partial VSD speed and motor brake-horsepower). ○ Pump curve shall show the performance indicated at the maximum operating point (flow and pressure, based on pump impeller size, full VSD speed, and full brake-horsepower). ○ PCW pumps shall be Cast Iron or Stainless Steel casing, with Stainless Steel internally wetted parts (shaft, impeller, etc.). • Pump motor sizing and selection shall be as follows: ○ Pump motors shall be selected for non-overloading operation on all points of the performance curve at the design conditions. ○ Pump motors shall be rated for inverter duty with variable speed operation. ○ Pump motor efficiency rating shall be minimum IE3. ○ Pump motor IP rating shall be minimum IP55. • Pump installation shall be as follows: ○ Each PCW pump shall be installed with full size isolation valves, pressure gauges, flexible connections, and low point drain valves on both the return and discharge side of the pump. ○ Each PCW pump shall have a strainer or suction diffuser installed on the pump suction side, and a check valve installed on the pump discharge side. ○ Each PCW pump shall have concrete inertia base and vibration isolation springs. ○ Each PCW pump shall be installed on minimum 4-inch (100mm) high concrete housekeeping pad.

Equipment	Type and Description
	Note: Approved Manufacturers – Goulds, KSB, Andritz, Ebara, or approved equivalent
Heat Exchangers	• Heat Exchangers shall be Plate and Frame Type. • Heat Exchanger sizing and selection shall be as follows: ○ Heat Exchangers shall be sized for 4.5 to 5.5 °C (~8.0 to 10.0 °F) ΔT on the PCW side. **Note**: During actual system operation, the actual ΔT on PCW side may typically be ~3.0 to 3.5 °C (~5.4 to 6.3 °F) ΔT. ○ Each Heat Exchanger shall have expansion capability to add additional plates in the future. The expansion capability shall allow for a future increase in capacity by a minimum of 20%-50% capacity. ○ Heat Exchangers shall be Carbon Steel frame with Stainless Steel plates, Stainless Steel wetted parts/piping connections, and EPDM gaskets. • Heat Exchanger installation shall be as follows: ○ Each Heat Exchanger shall be installed with temperature gauges and pressure gauges on both the supply and return side of the Heat Exchanger for hot side and cold side. ○ Each Heat Exchanger Chilled Water control valve shall have a partial-line size bypass with isolation valve, around the control valve. This is to allow maintenance and repair to be performed on the control valve, while keeping the Heat Exchanger in manual (bypass) operation. ○ Provide stainless steel containment drip pan below the heat exchangers and flange connections.
Filters	• Filters shall be horizontal configuration. This is to allow easy access for maintenance and filter replacement. • Filters shall be cartridge type with minimum rating of 50 microns particulate size. • Filters shall be Stainless Steel shells and housings. • Each Filter shall be installed with air vent port, drain port, and differential pressure gauge.

Equipment	Type and Description
	• A full line-size bypass with isolation valve shall be installed around the Filters for use during startup or for other needs (minimum size to match 1 Filter housing).
Air Separator(s)	• Air Separator shall be Tangential, Cyclone Type. • Air Separator shall be constructed of Stainless Steel shells and housings. • A full line-size bypass with isolation valve shall be installed around the Air Separator for use during startup, maintenance, or repair.
Expansion Tank(s)	• Conventional expansion Tank shall be Diaphragm or Bladder Type. • Expansion Tank shall be provided with drain port, air charging fitting, pressure gauge, and pressure relief valve. • Expansion tank shall not be used as makeup water control.
Water Quality and Chemical Dosing Skid	• Manual Pot Feeder or Automated Metering Pumps shall be provided to allow dosing of chemicals. • Chemical dosing provisions shall include corrosion inhibitor and biocide, and provisions may also include pH adjustment and conductivity adjustment. • Chemical Corrosion Rack shall be provided to allow monitoring of corrosion rates across different materials.

8.0 Piping Design and Sizing Requirements:

8.1 **Piping Diversity and Sizing Factor** – The PCW main piping shall be sized according to N+2F the System Diversity and Sizing factors stated in the above sections.

8.2 Each PCW lateral shall be sized so that the lateral's capacity meets 115-120% of the PCW demand at full build for the lateral (after initial construction of the system), based on the manufacturing tool layout and TUM. Therefore, each PCW lateral shall have 15-20% spare future capacity over the PCW demand at full build for the lateral (after initial construction of the system). See **Table 5**.

Table 5: PCW Pipe Sizing

- PCW Main Piping Sizing = N + 2F
- PCW Lateral Demand at Full Build, DL = TUM Peak Demand for Lateral x 100% Diversity
- PCW Lateral Piping Sizing = 115-120% x D_L

8.3 **Pipe Sizing –** PCW Pipe Sizing shall be selected based on the full build capacity of the PCW system equipment, including the future equipment (N+F). The Pipe Sizing shall be designed with maximum flowrate not to exceed the values in **Table 6**.

Table 6: Pipe Sizing Criteria

PCW Nominal Pipe Size	Maximum Flowrate			Approximate Velocity
	GPM	CMH	LPM	
2″ DN50	60	14	233	6.6 fps (2.0 m/s)
2.5″ DN65	105	24	400	6.6 fps (2.0 m/s)
3″ DN80	198	45	750	8.2 fps (2.5 m/s)
4″ DN100	308	70	1,167	8.2 fps (2.5 m/s)
6″ DN150	700	159	2,650	8.2 fps (2.5 m/s)
8″ DN200	1,250	283	4,716	8.2 fps (2.5 m/s)
10″ DN250	2,140	486	8,100	9.0 fps (2.75 m/s)
12″ DN300	3,082	700	11,667	9.0 fps (2.75 m/s)
14″ DN350	4,195	953	15,884	9.0 fps (2.75 m/s)
16″ DN400	5,477	1,244	20,733	9.0 fps (2.75 m/s)
18″ DN450	6,930	1,574	26,233	9.0 fps (2.75 m/s)
20″ DN500	8,554	1,943	32,383	9.0 fps (2.75 m/s)
22″ DN550	10,544	2,395	39,916	9.2 fps (2.8 m/s)
24″ DN600	12,548	2,850	47,500	9.2 fps (2.8 m/s)
26″ DN650	14,727	3,345	55,750	9.2 fps (2.8 m/s)
28″ DN700	17,079	3,879	64,650	9.2 fps (2.8 m/s)
30″ DN750	19,605	4,453	74,215	9.2 fps (2.8 m/s)

8.4 Materials of Construction - Distribution piping shall be 304L Schedule 10 Stainless Steel.

8.4.1 Piping sections shall be joined via welded ends as much as possible, to minimize the number of flanges and mechanical joints in the system.

8.5 **Spare Valves for Main Expansion** – Full size isolation valves shall be provided at the ends of all main piping headers for equipment (i.e. main piping headers for PCW Pump Suction and Discharge, Heat Exchanger Inlet and Discharge, Filter Inlet and Discharge, and CHW Supply and Return). The future valves are to allow future capacity expansion or cross-connections, etc.

8.6 **Spare Valves for Future Laterals** - Additional isolation valves shall be included on the main supply and return piping distribution piping, to allow installation of future laterals for capacity expansion. Based on the number of laterals included in the design, a minimum quantity of 35-50% of additional isolation valves for future laterals shall be included.

8.7 **Lateral Point-of-Connection Quantity and Sizes** – The laterals shall have sufficient quantity of Point of Connections (POCs) to meet the tool layout and requirements.

8.7.1 A minimum quantity of 30-40% of additional spare POCs shall be provided on each lateral, for future tools.

8.7.2 The sizes for each POC shall be checked and verified versus the actual tool requirements. Minimum POC size shall be DN65 (2-½”) size, but larger POC sizes can be provided, based on the actual tool requirements.

8.8 **Lateral End of line Bypass Valves -** Provide lateral end-of-line bypass piping and isolation valves on all PCW Supply and Return laterals for use during startup and low load conditions. For startup, the lateral end-of-line bypass piping shall be minimum size of DN50 (2”); this bypass piping can be achieved via temporary piping connected across the LPOCs, if it is preferred.

8.9 **Makeup-Water Piping –** UPW/DI/RO Water shall be used as the normal makeup water source. Makeup Water piping shall be provided with check valves/backflow preventer, pressure regulating valve, pressure gauges, and modulating-type automatic control valve.

8.10 **Emergency Quick-Fill Makeup-Water Piping -** ICW Water shall be used as the emergency quick-fill makeup water source. Makeup Water piping shall be minimum DN65 (2-½") size and shall be provided with check valves/backflow preventer, pressure regulating valve, pressure gauges, and modulating-type automatic control valve.

8.11 **Control Valve Bypass Piping -** Bypass piping with isolation valves shall be installed across the UPW/DI/RO Makeup Water Control Valve, ICW Quick-Fill Makeup Water Control Valve, Conductivity Blow-Down Control Valve, and each Heat Exchanger Chilled Water Control valve, to allow future isolation for repair or replacement of the valves.

9.0 Accessories:

9.1 The following accessories which shall be installed – See **Table 7**.

Table 7: Accessories to be installed

Accessories	Type and Installation Location	Remarks
Piping Insulation	• Install for all piping in PCW mechanical room spaces, corridors, pipe risers, non-subfab/fab spaces, unconditioned spaces, or other spaces where condensation is possible	In accordance with Spec Section 15260 (or equivalent Spec)
Piping Supports	• Install piping hangers, supports, racks, and clamps at the appropriate spacing interval as required by the system and piping load and layout • Provide seismic restraints and supports as deemed required during design, based on site location and local jurisdiction requirements • Provide vibration isolation for piping supports as deemed required during design, based on the building vibration criteria	In accordance with Spec Section 15140 (or equivalent Spec)
Flexible Connections	• Install at all pump connections on suction and discharge	
Pressure Gauges	• Install at each pump suction and discharge • Install at each heat exchanger inlet and outlet on both hot side and cold side • Install at lateral end of line (on both supply and return side)	

Accessories	Type and Installation Location	Remarks
	• Install Differential Pressure gauge on each filter	
Temperature Gauges	• Install at each Heat Exchanger inlet and outlet on both the PCW side and the CHW side	
Air Vents	• Air vents shall be minimum size DN20 (3/4 inches) • Install Auto or Manual Air Vents at all high point locations • Install Auto Air Vents at vertical piping connected to the pump suction • Install Auto Air Vents on the horizontal piping of main return line immediately before the pump suction • Install Auto or Manual Air Vents at all lateral end-of-line (on both supply and return ends)	
Drain Ports	• Install at all low point locations • Install at all pumps and heat exchangers • Install at all filter housings	
Valves	• All butterfly valves shall be rated for “bi-directional” duty with positive shutoff in both directions • All butterfly valves shall be lug-style	In accordance with Spec Section 15101 (or equivalent Spec)
Control Valves	• Provide all control valves with manual override function. • The Heat Exchanger Chilled Water Control valve actuators and positioners shall provide position feedback status of the actual valve position • Pneumatic type modulating valves shall be used for makeup water and blowdown lines.	
Labeling and Identification	• “PCWS” and “PCWR” nomenclature shall be used for Process Cooling Water Supply and Return • Provide identification labels at minimum of every 6m distance and change of direction	In accordance with Spec Section 15190 (or equivalent Spec)

Accessories	Type and Installation Location	Remarks
	• Labels shall be Green Background with White Lettering	

10.0 Electrical Design:

10.1 **Power Segregation** – Electrical power supply for PCW Pumps shall be segregated among a minimum of 2 separate upstream power sources (including panel + transformer).

10.2 **Emergency Power for Pumps** - A minimum of 50% of the PCW pumps shall be connected to Emergency Power, to continue running upon a loss of Normal Power.

10.3 **Emergency Power for Pumps** - 100% of PCW pumps shall be connected to an Un-interrupted Power source (such as UPS or DC Bank) to allow continuous operation and ride-through capability of the PCW pumps during voltage sags, spikes, and fluctuation events.

10.4 **Un-interrupted Power for Controls –** 100% of PCW system controls including PLCs, Local Control Panels (LCPs), and other auxiliary control power shall be connected to an Un-interrupted Power source (such as UPS or DC Bank) to allow continuous operation of the PCW system during momentary power interruption or voltage sags, spikes, and fluctuation events.

10.5 **Variable Frequency Drives –** All pumps shall be VFD driven for variable speed operation. Variable Frequency Drives shall be selected to be one size larger than the nameplate power rating of the pump motor size.

10.5.1 Variable Frequency Drives shall be compliant with SEMI standard F47 for voltage sag immunity, to provide ride-through capability during momentary power interruption or voltage sag.

10.6 **Local Disconnect Switches –** Each pump shall have a lockable type, local disconnect switch installed adjacent to the pump, to allow for manual isolation and lockout/tagout (LOTO/COHE) of the pump.

11.0 Instrumentation:

11.1 The following Instrumentation shall be installed and monitored and controlled via FMCS. See **Table 8**.

Table 8: Instrumentation to be installed

Instrumentation	Type and Installation Location
Pressure Transmitters	• Install at Main Supply Header after Final Filters (Quantity: 2 total, for 1 duty and 1 backup) • Install at Lateral end-of-line on both Supply and Return for a minimum of 2 laterals (Quantity: 4 total, for 2 supply and 2 return)
Temperature Transmitters	• Install at Main Supply Header after Final Filters (Quantity:2 total, for 1 duty and 1 backup)
Conductivity Meter	• Install at Sidestream loop for water sampling and chemical dosing (Quantity: 1 duty)
Supply Flowmeters	• Install at Main Supply Header after Final Filters (Quantity: 1 duty)
Makeup Water Flowmeters	• Install at UPW/DI/RO Make-up water piping • Install at ICW Emergency Quick-Fill Make-up water piping
Additional Spare Ports	• All supply and return laterals shall have a spare port and spare valve installed at the lateral end-of-line, to allow for future pressure transmitters to be installed. The future pressure transmitters will allow continuous SCADA monitoring of each lateral pressure.

12.0 Control Sequence Philosophy:

12.1 The overall philosophy for system control and operation shall be listed in this section. See **Table 9**. Refer to the specific Sequence of Operations for each project for specific details.

Table 9: System Control Logic

Control Parameter	Control Logic
Pressure Control	• Speed output of Pump VSDs shall be modulated to maintain the supply header pressure setpoint (as measured by main pressure transmitters, located downstream of final filters) • Automatic blow-down control valve shall open for blow-down when return pressure raises above high pressure

Control Parameter	Control Logic
	limit/setpoint. (As measured by main PIT, located at the return main header.)
Heat Exchanger Temperature Control	• Position output of Heat Exchanger control valves shall be modulated to maintain the supply header temperature setpoint (as measured by main temperature transmitters, located downstream of heat exchangers)
Conductivity Control	• Automatic Blow-down control valve shall open for blow-down when conductivity level raises above the high level setpoint (as measured by main conductivity transmitter, located at the water quality/chemical dosing skid)
Makeup Water Control	• Automatic UPW/DI/RO makeup water control valve shall open for make-up when return header pressure falls below the low pressure setpoint (as measured by main pressure transmitters, located at the main PCW return piping). The manual Pressure Regulating Valve on the UPW/DI/RO makeup line shall be set to limit the increase of system pressure when the control valve opens. • Automatic ICW Quick-Fill makeup water control valve shall open for make-up when the return header pressure falls below the low-low pressure setpoint. The manual Pressure Regulating Valve on the ICW Quick-Fill makeup line shall be set to limit the increase of system pressure when this control valve opens.
Pump Start/Stop	• Pump Start/Stop shall be done automatically via SCADA based on required number of running pumps

12.2 The following table shows the control valve fail positions that shall be configured. See **Table 10.**

Table 10: Control Valve Fail Positions

Control Valve	Control Valve Fail Position		
	Loss of Air	**Loss of Electricity**	**Loss of Control Signal**
Heat Exchanger Temperature Control Valve	Fail Last State	Fail Last State	Fail Last State
UPW/DI Makeup Water Control Valve	Fail Closed	Fail Closed	Fail Closed

Control Valve	Control Valve Fail Position		
	Loss of Air	Loss of Electricity	Loss of Control Signal
ICW Quick-Fill Makeup Water Control Valve	Fail Closed	Fail Closed	Fail Closed
Conductivity Blow-Down Control Valve	Fail Closed	Fail Closed	Fail Closed

12.3 Any control valve actuators configured to fail-last state shall have the capability of failing to last state position and holding the position for unlimited duration without needing control air, power, or PLC signal.

12.4 Upon a loss event of control air, power, or PLC signal, and then after the utility is restored, the control valve output command/position shall be set to the same command/position it was operating at prior to the loss event.

12.5 All control valves shall have a manual override bypass, to allow manual operation of the valve upon a failure.

13.0 Smart Facilities Design:

13.1 The following are additional enhancements that can be considered and included to enable Smart Facilities Design concepts.

13.1.1 **Lateral Pressure Online Monitoring –** Consider installing pressure transmitters at all PCW supply and return laterals at each end-of-line location, to allow continuous SCADA monitoring and alarm of each lateral pressure.

13.1.2 **Load Demand Online Monitoring** – Consider installing flowmeters at each building/area or each major load/user, to allow continuous SCADA monitoring and tracking of PCW load and demand each different user.

13.1.3 **Pump Vibration Online Monitoring** – Consider installing vibration sensors on each Pump and motor, to allow continuous monitoring and alarm of pump vibration levels.

14.0 Inter-Discipline Coordination Matrix:

14.1 The following table shows the provisions that shall be provided by each of the following disciplines and coordinated accordingly. See **Table 11**.

Table 11: Provisions Provided by Each Discipline

Discipline	Provision
Architectural	• Provide dedicated Utility Room or Mechanical Room space for centralized Process Cooling Water equipment • Provide concrete housekeeping pad below all equipment including pumps, heat exchanger, filters, air separator, expansion tank, etc. • Cut openings in walls and floors for piping penetrations, and seal openings after piping installation • Provide adequate fire stopping at all required fire wall penetrations
Mechanical	• Provide HVAC space conditioning for dedicated Utility Room or Mechanical Room space containing the centralized Process Cooling Water equipment. • Provide chilled water piping and accessories as outlined in this Standard • Provide floor drains located suitably to allow proper draining of equipment, piping, and air vents
Electrical	• Provide VFDs for all pumps • Provide local disconnect switches for all pumps • Provide power wiring for all pumps and Electrical equipment/components • Provide grounding for all pumps and Electrical equipment/components • Provide Emergency Power / UPS Power as outlined in this Standard • Provide Power Segregation as outlined in this Standard
Water / Wastewater	• Provide UPW/DI/RO Makeup water piping • Provide ICW Quick-Fill Makeup water piping • Provide Wastewater drain piping for blowdown
Controls	• Provide control valve actuators and positioners • Provide control wiring • Provide instrument control air tubing • Provide FMCS/SCADA system monitoring and controls of all instrumentation and transmitters including pressure, temperature, flow, conductivity, etc.

15.0 Quality Inspection and Testing:

15.1 **Inspection and Test Plan (ITP) –** Create formal Inspection and Test Plans (ITP) that outline all requirements for quality verification, inspections, testing, and commissioning:

15.1.1 Piping

- Piping Shop Drawings and Submittals
- Incoming Material Inspection
- Material Storage
- Welding Procedures, Qualifications, and Inspection
- Piping and Supports Installation and Verification
- QA/QC Inspection & Punchlist
- Leak Testing and Pressure Testing
- Cleaning and Flushing
- Quality Certification
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

15.1.2 Equipment

- Equipment Shop Drawings and Submittals
- Factory Acceptance Test
- Incoming Equipment Inspection
- Equipment Move-In and Installation
- QA/QC Inspection & Punchlist
- Equipment Pre-Startup Checks – Mechanical, Electrical, Controls, etc.
- Operational Acceptance Test (OAT) - Standalone Equipment Startup per Vendor’s Procedure to allow equipment to run in MANUAL mode. Also considered “Early Run” Equipment.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

15.1.3 System

- Site Acceptance Test (SAT) – Test System operation and control functions in accordance with P&ID, SOO, and Specifications.
- Integrated System Test (IST) – Test full System operation including all system interdependencies, interlocks, and critical failure modes.
- Final QA/QC Inspection & Punchlist
- Training, Documentation, and Handover

15.2 **QA/QC Inspection** – Prior to leak testing and pressure testing, QA/QC inspection shall be performed on all installed piping and components to verify proper installation in accordance with construction drawings, project specifications, and industry standards.

15.3 **Gross Leak Test** - Gross leak test shall be performed via compressed Nitrogen or Air at pressure up to 5 psig. Walk the entire system to verify that no leaks are present, and to make repairs to any leaks that are identified.

15.4 **Hydrostatic Pressure Test** - All piping shall be hydronic pressure tested at a minimum of 1.5 times the design operating pressure, for a minimum of 2-4 hours. Properly vent the system to remove air via all air vents, during the water filling process.

15.5 **Cleaning and Flushing -** Circulate a mixture of clean water and appropriate cleaning solution for a minimum of 12-24 hours at a minimum velocity of 2.5-3.0 fps (0.6-0.9 m/s). Properly vent the system to remove air via all air vents, during filling process. Drain the water after completion of flushing process.

15.6 **Final Fill –** Fill the system with UPW/DI/RO water, and introduce chemical treatment immediately to bring the water quality within spec. Properly vent the system to remove air via all air vents, during filling process. Obtain approval from Site Facilities Operations team on the final water quality.

16.0 Startup Commissioning / Site Acceptance Test

16.1 At a minimum, the following system control functions shall be tested and successfully proven during the System Startup Commissioning / Site Acceptance Test.

16.1.1 Pumps

- Verify Standalone Commissioning for each Pump, per manufacturer's instructions
- Verify Pump Operational Function through the operating range (0 to 50/60Hz)
- Verify Pump Start/Stop Function to Rotate Operating Pumps
- Verify Pump Auto-Start sequencing, upon failure of an operating Pump
- Verify Pump Speed control in response to PCW pressure fluctuations
- Confirm PCW system pressure control is stable during pump manual start/stop rotation
- Confirm PCW system pressure control is stable during pump fault/trip event and pump auto-start sequencing

16.1.2 Heat Exchanger

- Verify Standalone Commissioning for each CCW Heat Exchanger, per manufacturer's instructions
- Verify Heat Exchanger CHW Control Valve Function through the operating range (0 to 100%)
- Confirm Heat Exchanger CHW Control Valve control in response to CCW supply temperature fluctuations

16.1.3 Makeup Water and Blowdown Drain

- Verify Automatic UPW/DI/RO makeup water control valve and Automatic ICW Quick-Fill makeup water control valve function in response to low pressure signal
- Confirm system pressure control is stable during UPW/DI/RO makeup water control valve and Automatic ICW Quick-Fill makeup water control valve function, in response to low pressure signal or loss of system water
- Verify Automatic blow-down control valve function in response to high conductivity level signal

16.1.4 Instrumentation and Controls

- Confirm Auto-switch to backup pressure transmitter upon failure of main pressure transmitter
- Confirm Auto-switch to backup temperature transmitter upon failure of main temperature transmitter
- Confirm all control valves fail to the stated position per this Standard, upon loss of instrument air, power, and/or control signal.
- Verify all alarms are enabled and activated in SCADA

17.0 Reference Specifications:

17.1 The following Construction Specifications shall be referenced during design, engineering, and construction. The Specifications shown below are in accordance with the Construction Specifications Institute (CSI) Master Format; Refer to the specific Construction Specifications for each project. See **Table 12**.

Table 12: Reference Construction Specifications

Spec Section	Spec Contents	Remarks
Section 15461	Process Cooling Water System	
Section 15140	Piping Supports and Anchors	
Section 15190	Piping Labels and Identification	
Section 15260	Piping Insulation	
Section 15280	Equipment Insulation	
Section 15515	Hydronic Specialties	
Section 15990	Testing, Adjusting and Balancing	
Section 15170	Motors	
Section 15172	Variable Frequency Drives	
Section 15070	Stainless Steel Piping	
Section 15101	Valves	
Section 15540	Pumps	
Section 11223	Filters	
Section 15242	Vibration Isolation	
Section 15695	Steel Tanks	

18.0 Document Control

18.1 **Approval**: GLOBAL_FAC_GFTT_APPR

18.2 **Notification**:
GLOBAL_FAC_MANAGERS
GLOBAL_FAC_COMMUNICATIONS
GLOBAL_FAC_CONSTRUCTION
GLOBAL_FAC_NOTIFY
FCG_F4_FAC_NOTIFY
FCG_F6_FAC_NOTIFY
FCG_F10_FAC_NOTIFY
FCG_F11_FAC_NOTIFY
FCG_F15_FAC_NOTIFY
FCG_F16_FAC_NOTIFY
FCG_MMY_FAC_NOTIFY
FCG_MMP_FAC_NOTIFY
FCG_MSB_FAC_NOTIFY
FCG_MTB_FAC_NOTIFY
FCG_MXA_FAC_NOTIFY

18.3 **Final Viewer Access**
All Micron sites Facilities Team Members

18.4 **Retention**
Adherence to the requirements specified in the
Global Facilities - Document Control Procedures

18.5 **Review**
Biennially (two years)

19.0 Revision History

Revision #	Section Changed	Details of Changes	Date
0	NA	New document	03/27/2020
1	Section 6	Section 6.2 Added PCW system water quality specification Added “Table 2: Water Quality Specs for PCW”	04/01/2022
	Section 7	section 7.8 Table 4 Expansion Tank(s) Revised from “Expansion Tank...” to “Conventional expansion tank...” Added “Expansion tank shall not be used as makeup water control”.	
	Section 9	9.1 Table 7 “Control Valves” add "Pneumatic type modulating valves shall be used for makeup water and blowdown lines".	
	Section 12	Section 12.1 Table 9 “Pressure Control” Revised from “Pump Pressure Control” to “Pressure Control” Added automatic blow-down control logic when return pressure raises above high-pressure limit/setpoint. (As measured by main PIT, located at the return main header.)	

Appendix 1: Template Form for Site Specific Deviation List

Site:______________________

No	Specification Section and Item	Deviation Proposal	Scope Impact (Benefit, Cost, Schedule)	Remark	Approval: 1. Site 2. GFTT